

Drowning & Submersion Injuries

Guideline Title: Drowning & Submersion Injuries

Guideline Number: GUID-3203-T010

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CXVII. **PURPOSE:** To establish and maintain adequate airway, breathing, circulation, oxygenation, and ventilation and to recognize and treat associated trauma and hypothermia. Drowning refers to “a process resulting in primary respiratory impairment from submersion or immersion in a liquid medium” (Utstein). Conditions associated with drowning include alcohol and drug use, seizures, and traumatic injuries. The temperature of the water, duration of submersion, cleanliness of water, and age of the patient all influence the outcome after submersion injuries and are thus important details to obtain at the scene. Other factors that influence outcome include the duration of cardiac arrest and time to initiation of resuscitation. Hypoxemia is the most life-threatening consequence and hypovolemia is a common late onset problem secondary to sequestration of fluid in alveoli and ARDS.

CXVIII. **DEFINITIONS:**

CXIX. **DEPARTMENT GUIDELINE:**

- A. Establish care as defined by *GUID-G002 – General Patient Care*.
- B. Consider the possibility of trauma (visible trauma/diving accident/shallow water/axial loading).
 - i. Spinal precautions should be initiated and maintained if indicated.
 - ii. Pediatric patients <16 years old should be transported to a pediatric trauma center.
- C. Assess and manage airway, breathing, and circulation.
 - i. Administer supplemental oxygen to maintain SpO₂ >93%.
 - ii. If actual or potential airway compromise, respiratory failure, poor ventilatory effort, need for ventilator assistance, or airway protection due to altered mental status or decreased level of consciousness, secure airway per *GUID-G008 – Rapid Sequence Intubation for Endotracheal Intubation* and *GUID-PR020 – Endotracheal Intubation*. If laryngoscopy or passage of an ET tube is not possible follow *GUID-PR016 – King LTS-D Laryngeal Tube*.
 - iii. If patient is intubated, appropriate ventilation should be initiated per *GUID-PR024 – Ventilation with a Mechanical Ventilator* as soon as possible.
 1. Utilize PEEP. Start at 5cm H₂O & observe for improvements in oxygen saturations.
 2. Consider low tidal-volume lung protective mechanical ventilation.

3. If signs of neurological deterioration exist, maintain ETCO₂ between 30-35mmHg.
 - iv. Monitor ETCO₂ per *GUID-PR009 – End-Tidal CO₂ Monitoring*.
 - v. If bronchospasm is present, administer **Albuterol 2.5mg in 3ml of NS via nebulizer**. Repeat as necessary per *GUID-PR012 – Inhalation Therapy*.
 - vi. Consider obtaining a 12-lead per *GUID-PR001 – 12-Lead Electrocardiogram (ECG/EKG)*.
 1. Treat all cardiac dysrhythmias per Cardiac Guidelines.
- D. Obtain IV access per *GUID-PR019 – Intravenous Insertion*.
- E. If signs of inadequate perfusion, begin resuscitation per *GUID-M011 – Hypotension or GUID-PED009 – Pediatric Hypotension*.
- F. Perform a glucose check and if glucose is <60 mg/dL, and the patient is symptomatic, administer medications per *GUID-M021 Hypoglycemia*.
- G. If intubated, place gastric tube to decompress stomach contents per *GUID-PR010 – Gastric Tube Insertion*.
- H. Remove all wet clothing. Quickly start to warm and dry the patient with blankets. Consider utilizing infant warming pad. Prevent further heat loss and monitor body temperature. Treat per *GUID-M012 – Hypothermia*. Consider warm IV fluids per *GUID-PR026 – Warming of Blood Products & Intravenous Fluids*.
- I. If a seizure is suspected, follow guidance per *GUID-M015 – Seizures*.
- J. For patients who remain unresponsive and do not respond to above measures, consider ingestions or intoxications.

CXX. **DOCUMENTATION:** EMS Charts

CXXI. **REFERENCES:**